

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

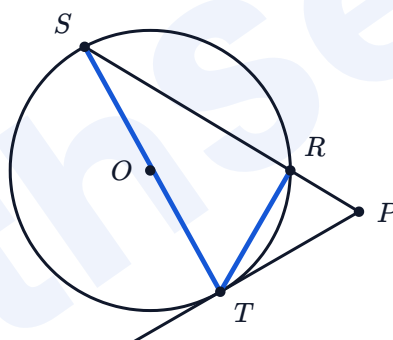
Chapter 18: Tangents and Intersecting Chords

Previous year board questions, topic-wise: 12 questions

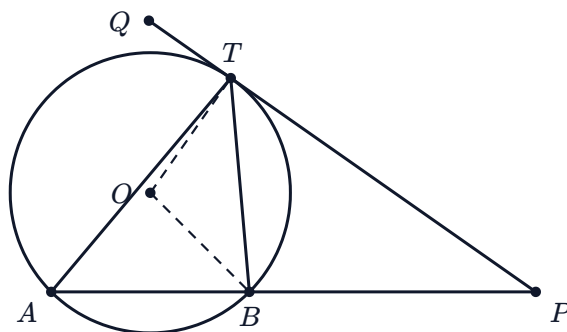
The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Tangent and secant from an external point

- PT is a tangent to a circle at T , and a line through P cuts the circle at R and S , with R nearer to P . (a) Prove that $\triangle PTR \sim \triangle PST$. (b) Prove that $PT^2 = PR \times PS$. (c) If $PR = 4$ cm and $PS = 16$ cm, find PT . **ICSE 2025, 3 marks**

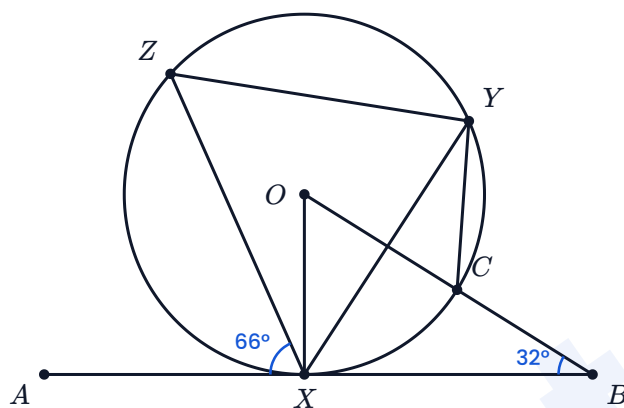


- A circle has centre O and radius 9 cm. The diameter AB is produced to a point P , and PQ is a tangent from P . If $PA = 24$ cm, find PQ . **ICSE 2024, 3 marks**
- PQ is a tangent to a circle with centre O at T . The chord AB produced meets the tangent at P . $AB = 9$ cm, $BP = 16$ cm, $\angle PTB = 50^\circ$ and $\angle OBA = 45^\circ$. Find (a) PT (b) $\angle BAT$ (c) $\angle BOT$ (d) $\angle ABT$. **ICSE 2023, 4 marks**

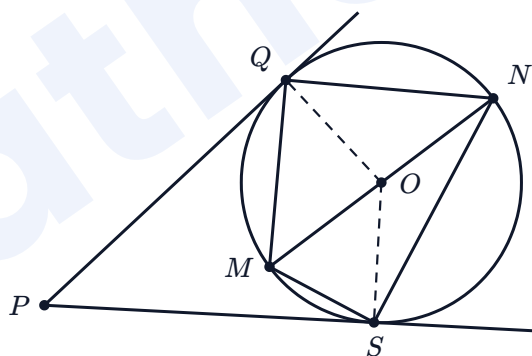


2 The alternate segment theorem

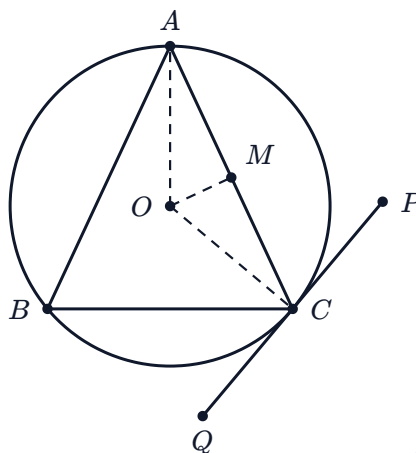
4. X, Y, Z and C lie on a circle with centre O . AB is the tangent at X , OB cuts the circle at C , and $ZY = XY$. Given $\angle OBX = 32^\circ$ and $\angle AXZ = 66^\circ$, find (a) $\angle BOX$ (b) $\angle CYX$ (c) $\angle ZYX$ (d) $\angle OXY$. **ICSE 2025, 4 marks**



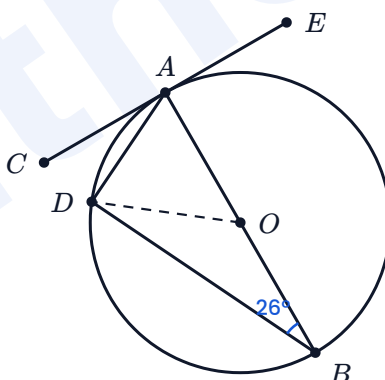
5. PR and PT are tangents from P to a circle with centre O , touching it at Q and S . MN is a diameter, with M on the arc QS nearer to P . Given $\angle PQM = 42^\circ$ and $\angle PSM = 25^\circ$, find (a) $\angle OQM$ (b) $\angle QNS$ (c) $\angle QOS$ (d) $\angle QMS$. **ICSE 2024, 4 marks**



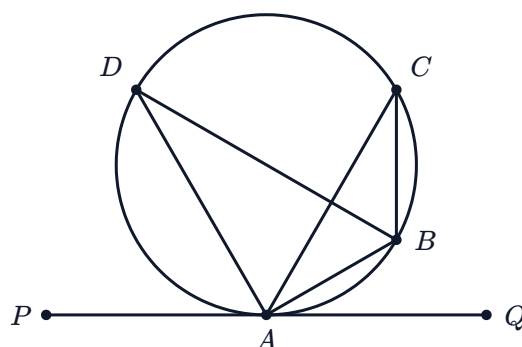
6. An isosceles triangle ABC with $AB = AC$ is inscribed in a circle with centre O . PQ is the tangent at C , with Q on the side of BC away from A . OM is perpendicular to the chord AC and $\angle COM = 65^\circ$. Find (a) $\angle ABC$ (b) $\angle BAC$ (c) $\angle BCQ$. **ICSE 2024, 3 marks**



7. RT is a tangent to a circle at S . P and Q are points on the circle with $\angle PST = 30^\circ$ and $\angle SPQ = 60^\circ$. Find $\angle PSQ$. (a) 40° (b) 30° (c) 60° (d) 90° **ICSE 2023, 1 mark**
8. AB is a diameter of a circle with centre O , and CE is the tangent at A . D is a point on the circle on the same side of AB as C , with $\angle ABD = 26^\circ$. Find (a) $\angle BDA$ (b) $\angle BAD$ (c) $\angle CAD$ (d) $\angle ODB$. **ICSE 2023, 4 marks**

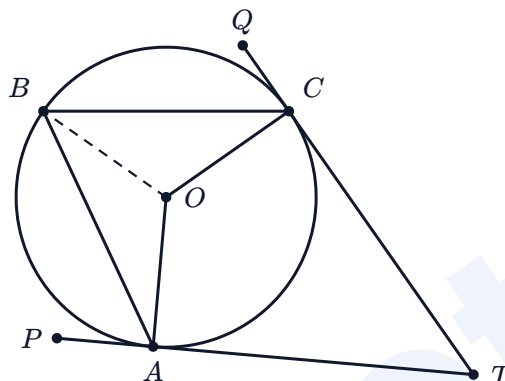


9. PQ is a tangent to a circle at A . B, C and D are points on the circle such that AB bisects $\angle CAQ$ and AD bisects $\angle PAC$. If $\angle BAQ = 30^\circ$, prove that (a) BD is a diameter (b) $\triangle ABC$ is isosceles. **ICSE 2017, 3 marks**

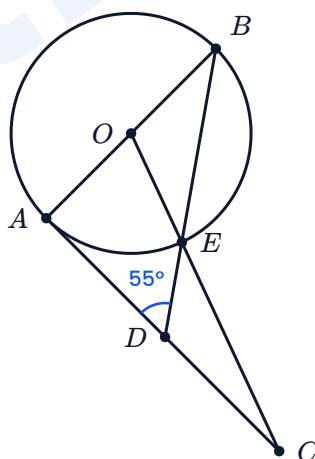


3 Tangent properties

10. PS and PT are tangents to a circle from P , with $\angle SPT = 80^\circ$. Q is a point on the circle with $SQ \parallel PT$. Find $\angle QST$. (a) 140° (b) 90° (c) 80° (d) 50° **ICSE 2024, 1 mark**
11. TP and TQ are tangents to a circle with centre O , touching it at A and C . B is a point on the circle with $\angle BCQ = 55^\circ$ and $\angle BAP = 60^\circ$. Find (a) $\angle OBA$ and $\angle OBC$ (b) $\angle AOC$ (c) $\angle ATC$. **ICSE 2020, 4 marks**



12. AB is a diameter of a circle with centre O , and AC is the tangent at A . D is a point on AC , and BD cuts the circle at E . The line OE produced meets the tangent at C . If $\angle ADB = 55^\circ$, find $x = \angle ACO$ and $y = \angle AOE$. **ICSE 2019, 3 marks**



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